UNIT-III: Network Layer

Switching, Logical addressing – IPV4, IPV6;
Address mapping – ARP, RARP, BOOTP, and DHCP–Delivery, Forwarding, and Unicast Routing protocols.

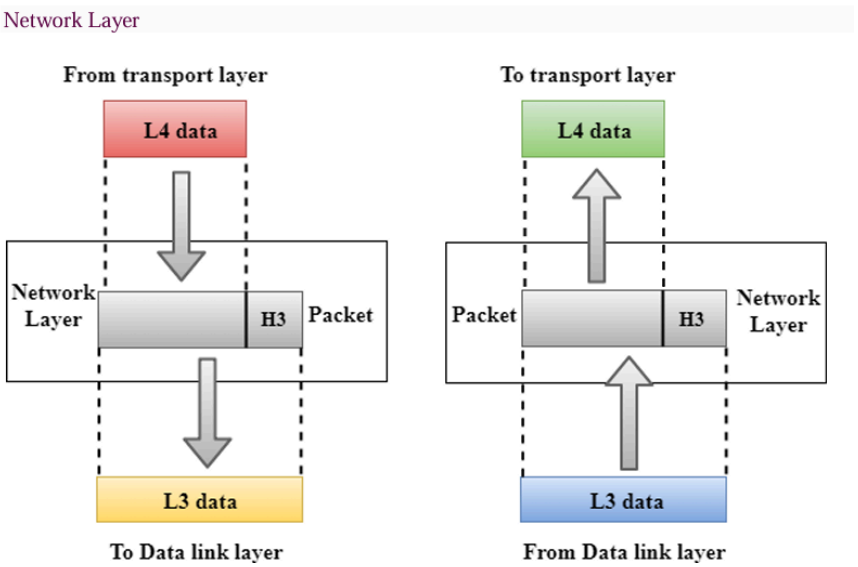
The **Network Layer** is Layer 3 of the OSI model and handles device addressing, routing, and data transfer across networks. Below is a summary of its key components and functions:

Key Points

- **Purpose:** Manages device addressing, tracks device locations on the network, and determines the best path for data transmission.
- **Routing and Forwarding:** Responsible for moving packets from source to destination.
- **Devices:** Routers, which are Layer 3 devices, provide routing services in an internetwork.
- **Protocols:** Common Network Layer protocols include:
    - IP (Internet Protocol)
    - IPv6

Functions of the Network Layer

1. **Internetworking:**
    - Provides a logical connection between different devices across networks.
2. **Addressing:**
    - Adds source and destination addresses to the header of frames for identification on the internet.
3. **Routing:**
    - Determines the best, most optimal path from source to destination out of multiple available paths.
4. **Packetizing:**
    - Converts data from the upper layers into packets through processes managed by protocols like IP.



Switching

Switching refers to the process of forwarding packets from an ingress port (incoming) to an egress port (outgoing) toward the destination. It involves various techniques to efficiently route data across a network.

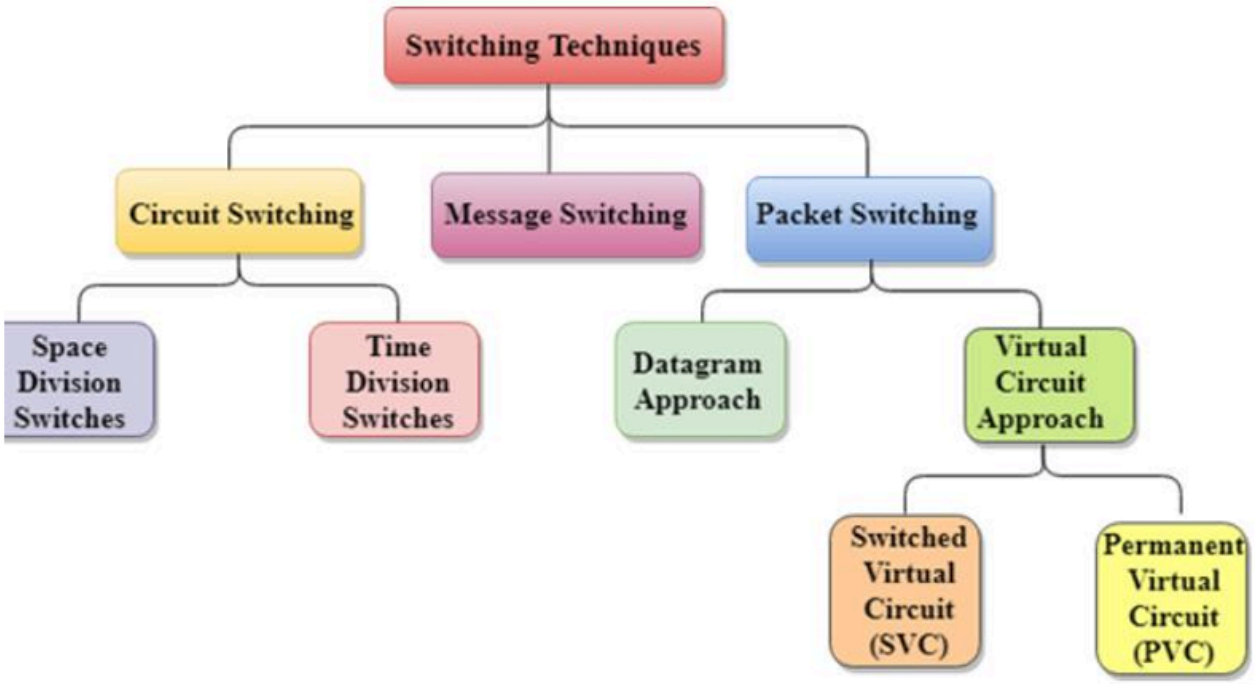
Types of Switching

1. Connectionless Switching

- **Definition:** Data is forwarded based on forwarding tables without pre-established connections.
- **Characteristics:**
    - No handshaking is required.
    - Acknowledgments are optional.

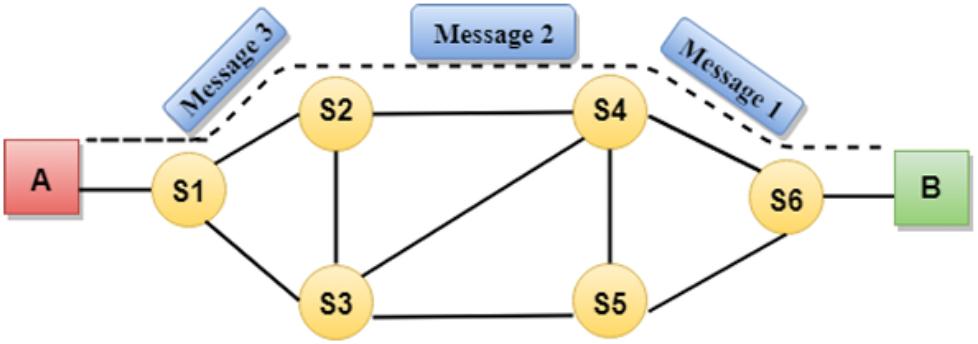
2. Connection-Oriented Switching

- **Definition:** Requires pre-establishment of a circuit between endpoints before data transfer.
- **Characteristics:**
    - Data is transmitted along a dedicated path.
    - Circuit is terminated after communication or reserved for future use.



Circuit Switching

- **Definition**: Establishes a dedicated path between sender and receiver before communication begins.
- **Key Features:**
  - End-to-end path exists before data transfer.
  - Commonly used in public telephone networks for voice transmission.
  - Fixed data transfer rate.



Phases of Circuit Switching:

1. **Circuit Establishment:** Connection is set up between endpoints.
2. **Data Transfer:** Communication occurs over the dedicated path.
3. **Circuit Disconnection:** Path is terminated after communication ends.

Circuit Switching Technologies:

- **Space Division Switches:**
  - Use crosspoints for transmission.
  - High-speed, non-blocking switches.
  - **Examples:** Crossbar Switch, Multistage Switch.
  - **Crossbar Switch:** High cost due to increased crosspoints.
  - **Multistage Switch:** Splits crossbar switches into smaller units, reducing cost and offering redundancy.

Advantages:

- Dedicated communication channel ensures reliability.
- Fixed bandwidth.

Disadvantages:

- Time-consuming to establish connections (~10 seconds).
- Inefficient for idle connections.
- High cost due to dedicated paths.

Message Switching

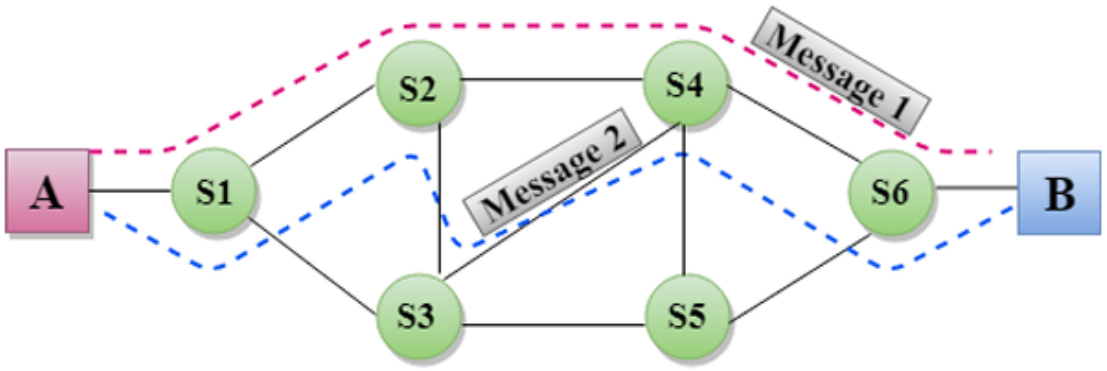
- **Definition:** Entire messages are routed through intermediate nodes using a store-and-forward mechanism.
- **Key Features:**
    - No dedicated path is required.
    - Intermediate nodes store and forward messages dynamically.
    - Supports messages of unlimited size.

Advantages:

- Efficient bandwidth usage.
- Reduces congestion with temporary message storage.
- Supports message prioritization.

Disadvantages:

- Requires significant storage at nodes.
- May cause delays due to storage and forwarding.



Packet Switching

- **Definition:** Divides messages into smaller packets, sent independently across the network.
- **Key Features:**
    - Packets contain headers with source, destination, and sequence information.
    - Reassembled in order at the destination.
    - Missing/corrupted packets are retransmitted.

Approaches:

1. **Datagram Packet Switching:**
    - Connectionless; each packet is treated independently.
    - Packets take varying routes.
2. **Virtual Circuit Switching:**
    - **C**onnection-oriented; a logical path is established before transmission.

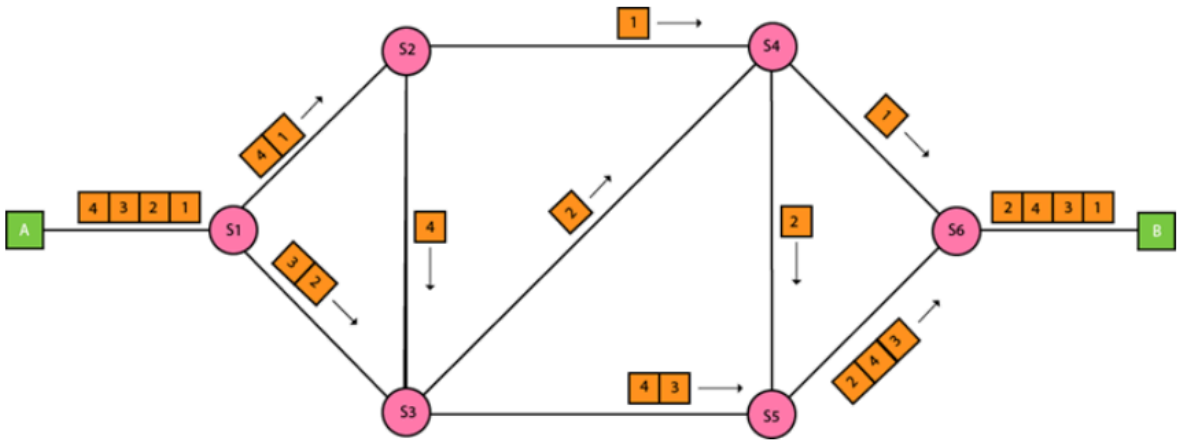
- Acknowledgment and termination signals are used.

Advantages:

- **Cost-effective:** Minimal storage requirements.
- **Reliable:** Packets reroute in case of busy nodes.
- **Efficient:** Shared communication channels optimize bandwidth usage.

Disadvantages:

- Not suitable for low-latency applications.
- Complex protocols with high implementation costs.
- Network congestion may lead to retransmissions and potential data loss.



Logical Addressing: IPv4 and IPv6

Logical addressing in networking refers to the unique identifiers assigned to devices for communication over a network. This is achieved through **IP Addresses**, which are governed by the Internet Protocol (IP). These addresses are provided by the Internet Protocol (IP) and fall under two main versions: IPv4 and IPv6.

IP Address

- **Definition**: A numerical label assigned to devices connected to a network for identification and communication.
- **Characteristics**:
  - Acts as an identifier for a specific device on a network.
  - Specifies the addressing format and packet routing scheme.
  - Commonly used with **Transmission Control Protocol (TCP)** to establish virtual connections.

Types of IP Addresses

1. **Static IP Address**:
    - Manually assigned and remains unchanged.
    - Typically more expensive than dynamic addresses.
    - Not always provided by all ISPs.
2. **Dynamic IP Address**:
    - Temporarily assigned and can change over time.
    - Managed using protocols like **DHCP (Dynamic Host Configuration Protocol)** or **PPPoE (Point-to-Point Protocol over Ethernet)**.

IPv4 Addressing

- **Format**: A 32-bit binary number divided into four octets separated by dots (e.g., 192.168.1.1).
- **Components**:
    - **Net ID**: Identifies the network.
    - **Host ID**: Identifies a device within the network.

Classful Addressing System

IPv4 addresses are divided into **five classes (A, B, C, D, E)** based on their leading bits.

1. **Class A**:
    - **First Bit**: 0
    - **Net ID**: 8 bits
    - **Host ID**: 24 bits
    - **Range**: 1–126
    - **Use**: Large networks (e.g., NASA).
    - Total Hosts: $2^{24} - 2 = 16,777,214$.
2. **Class B**:
    - **First Bits**: 10
    - **Net ID**: 16 bits
    - **Host ID**: 16 bits
    - **Range**: 128–191
    - **Use**: Medium-sized networks (e.g., banks).
    - Total Hosts: $2^{16} - 2 = 65,534$
3. **Class C**:
    - **First Bits**: 110
    - **Net ID**: 24 bits

- **Host ID**: 8 bits
- **Range**: 192–223
- **Use**: Small networks (e.g., colleges).
- Total Hosts: $2^8 - 2 = 254$.

4. **Class D**:
   - **First Bits**: 1110
   - **Purpose**: Reserved for **multicasting**.
   - **Range**: 224–239
5. **Class E**:
   - **First Bits**: 1111
   - **Purpose**: Reserved for future or experimental use.
   - **Range**: 240–255

Class of IP Address	Total Number of IP Addresses	1st Octet Decimal Range	Number of Networks available	Hosts per network	Default Subnet Mask
Class A	2^{31}	1 – 126	$2^7 - 2$	$2^{24} - 2$	255.0.0.0
Class B	2^{30}	128 – 191	2^{14}	$2^{16} - 2$	255.255.0.0
Class C	2^{29}	192 – 223	2^{21}	$2^8 - 2$	255.255.255.0
Class D	2^{28}	224 – 239	Not defined	Not defined	Not defined
Class E	2^{28}	240 – 254	Not defined	Not defined	Not defined

Classless Addressing

- **Definition**: Overcomes the limitations of classful addressing by allowing flexible allocation of IP ranges using **subnet masks**.
- **CIDR (Classless Inter-Domain Routing)**: Represents IP ranges (e.g., 192.168.1.0/24).

IPv6 Addressing

- **Format**: A 128-bit hexadecimal number divided into eight 16-bit blocks
- (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- **Purpose**: Address exhaustion in IPv4 led to IPv6 development.
- **Features**:
  - Vastly larger address space.
  - Improved routing efficiency and security.
  - Supports auto-configuration and mobility

Logical Addressing: IPv4 and IPv6 Overview

Logical addressing in computer networks involves assigning unique identifiers to devices within a network to facilitate communication. These addresses are provided by the Internet Protocol (IP) and fall under two main versions: IPv4 and IPv6.

IPv4 (Internet Protocol Version 4) Characteristics:

1. **Address Length**:
   - 32-bit address.
   - Written in dotted decimal notation (e.g., 192.168.0.1).
2. **Address Space**:
   - Provides approximately **4.3 billion addresses**.
   - Insufficient due to the exponential growth of devices.
3. **Address Classes**:
   - Divided into **Class A**, **Class B**, **Class C**, **Class D**, and **Class E**.
   - Classes define the range of addresses and usage (e.g., Class D for multicast, Class E for research).
4. **Structure**:
   - Divided into **Net ID** (identifies the network) and **Host ID** (identifies devices within the network).
5. **Protocols**:
   - Common protocols include **ICMP**, **ARP**, and **DHCP**.
6. **Limitations**:
   - Address exhaustion.
   - Lack of built-in security and scalability.

IPv6 (Internet Protocol Version 6)Characteristics:

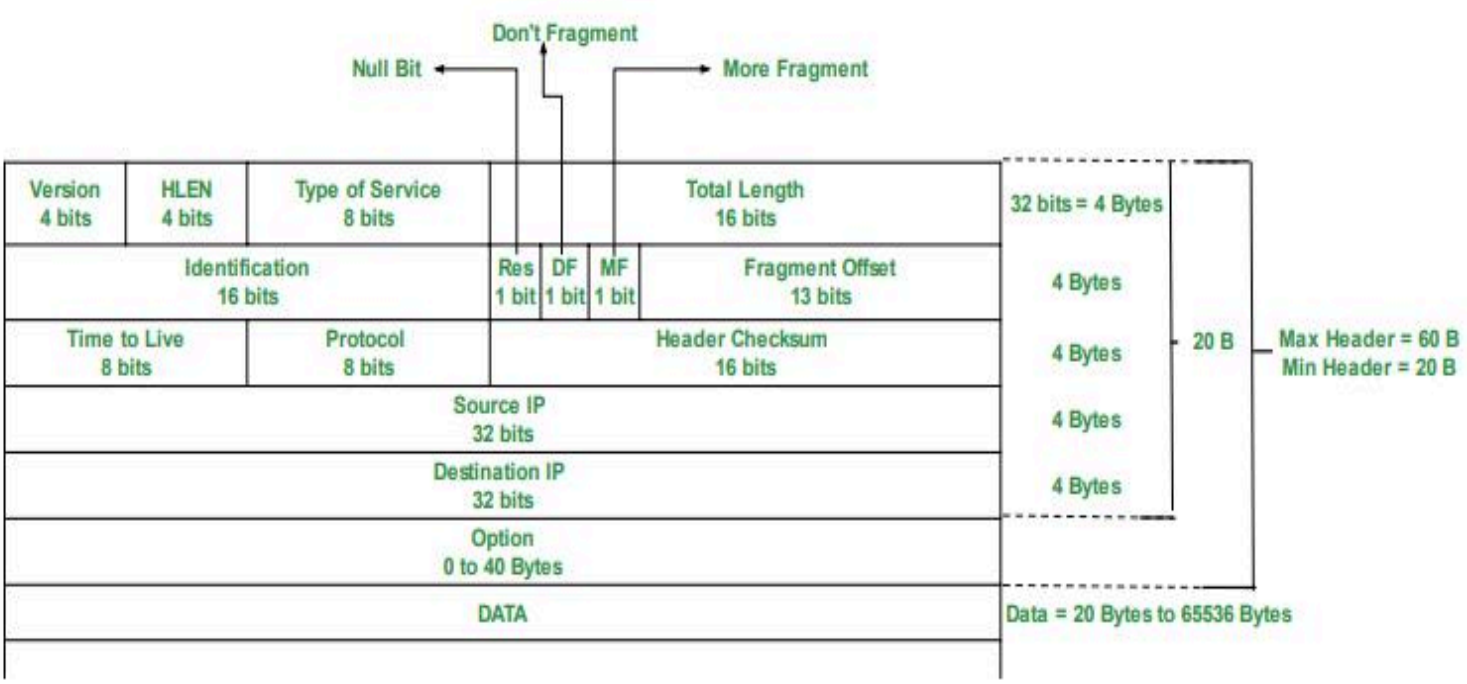
1. **Address Length**:
   - 128-bit address.
   - Written in hexadecimal and separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
2. **Address Space**:
   - Provides approximately **2^128** addresses, solving IPv4 exhaustion.
3. **Address Types**:
   - **Unicast**: For one-to-one communication.
   - **Multicast**: For one-to-many communication.
   - **Anycast**: For one-to-nearest communication.
4. **Structure**:
   - Divided into **Global Prefix**, **Subnet ID**, and **Interface ID**.
   - Supports hierarchical addressing.
5. **Key Features**:
   - Simplified header structure.
   - Auto-configuration (stateful and stateless).
   - Built-in security via IPSec.
6. **Advantages**:
   - Eliminates NAT due to abundant addresses.
   - Enhanced routing efficiency and QoS (Quality of Service).
7. **Protocols**:
   - ICMPv6, SLAAC, and DNSv6

IPv4 Header

The IPv4 header is more complex than IPv6 due to its fields related to fragmentation, checksums, and options. It is variable in length, ranging from **20 to 60 bytes**, depending on whether options are included. Below are the primary fields:

Key Fields:

1. **Version (4 bits)**: Specifies the IP version (value = 4 for IPv4).
2. **Header Length (4 bits)**: Indicates the length of the header in 4-byte blocks.
    - Minimum: 20 bytes (value = 5).
    - Maximum: 60 bytes (value = 15).
3. **Type of Service (8 bits)**: Used for Quality of Service (QoS).
4. **Total Length (16 bits)**: Total size of the packet, including header and data (max = 65,535 bytes).
5. **Identification (16 bits)**: Identifies fragments of the same packet.
6. **Flags (3 bits)**:
    - **DF (Don't Fragment)**: Prevents fragmentation.
    - **MF (More Fragments)**: Indicates additional fragments follow.
7. **Fragment Offset (13 bits)**: Indicates the position of a fragment in the original data.
8. **Time to Live (8 bits)**: Limits the packet's lifetime in hops.
9. **Protocol (8 bits)**: Indicates the transport layer protocol (e.g., TCP = 6, UDP = 17).
10. **Header Checksum (16 bits)**: Error-checking field for the header.
11. **Source IP Address (32 bits)**: IP address of the sender.
12. **Destination IP Address (32 bits)**: IP address of the receiver.
13. **Options (Variable)**: Optional parameters, e.g., route recording or source routing.

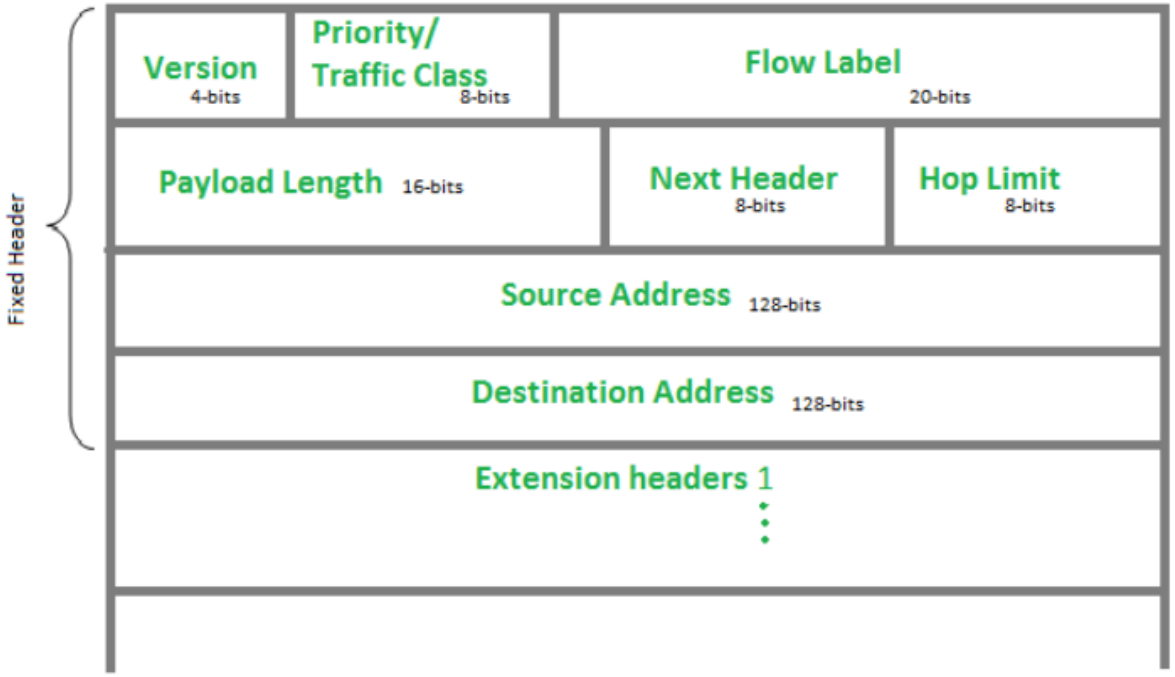


IPv6 Header

The IPv6 header is **fixed at 40 bytes**, making it simpler and more efficient than IPv4. Optional data is included in **extension headers**.

Key Fields:

1. **Version (4 bits)**: Specifies the IP version (value = 6 for IPv6).
2. **Traffic Class (8 bits)**: Used for QoS and Explicit Congestion Notification (ECN).
3. **Flow Label (20 bits)**: Identifies a specific flow for the packet stream.
4. **Payload Length (16 bits)**: Indicates the length of the payload (max = 65,535 bytes).
    - Set to 0 if the payload exceeds this size.
5. **Next Header (8 bits)**: Indicates the next protocol or extension header.
6. **Hop Limit (8 bits)**: Similar to TTL in IPv4; decrements at each hop.
7. **Source Address (128 bits)**: Address of the sender.
8. **Destination Address (128 bits)**: Address of the recipient.



IPV4 vs IPV6 :

IPv4	IPv6
IPv4 addresses are 32 bit length.	IPv6 addresses are 128 bit length.
IPv4 addresses are binary numbers represented in decimals.	IPv6 addresses are binary numbers represented in hexadecimals.
IPSec support is only optional.	Inbuilt IPSec support.
Fragmentation is done by sender and forwarding routers.	Fragmentation is done only by sender.
No packet flow identification.	Packet flow identification is available within the IPv6 header using the Flow Label field.
Checksum field is available in IPv4 header	No checksum field in IPv6 header.
Options fields are available in IPv4 header.	No option fields, but IPv6 Extension headers are available.
Address Resolution Protocol (ARP) is available to map IPv4 addresses to MAC addresses.	Address Resolution Protocol (ARP) is replaced with a function of Neighbor Discovery Protocol (NDP).
Internet Group Management Protocol (IGMP) is used to manage multicast group membership.	IGMP is replaced with Multicast Listener Discovery (MLD) messages.
Broadcast messages are available.	Broadcast messages are not available. Instead a link-local scope "All nodes" multicast IPv6 address (FF02::1) is used for broadcast similar functionality.
Manual configuration (Static) of IPv4 addresses or DHCP (Dynamic configuration) is required to configure IPv4 addresses.	Auto-configuration of addresses is available.

Address Mapping is the process of associating a logical address (such as an IP address) with a physical address (like a MAC address) in a network. Since devices communicate using both types of addresses, mapping ensures data packets reach their correct destination.

- **Logical Address**: This refers to network-layer addresses, like IP addresses, used to identify devices in a network.
- **Physical Address**: This refers to data-link layer addresses, like MAC addresses, used to identify devices at the hardware level (e.g., on the NIC).

Address mapping can be done in two ways:

1. **Static Mapping**: A table is manually created to associate logical addresses with physical addresses and stored on each machine.
2. **Dynamic Mapping**: The protocol (like ARP) is used to dynamically map the logical address to the corresponding physical address when needed.

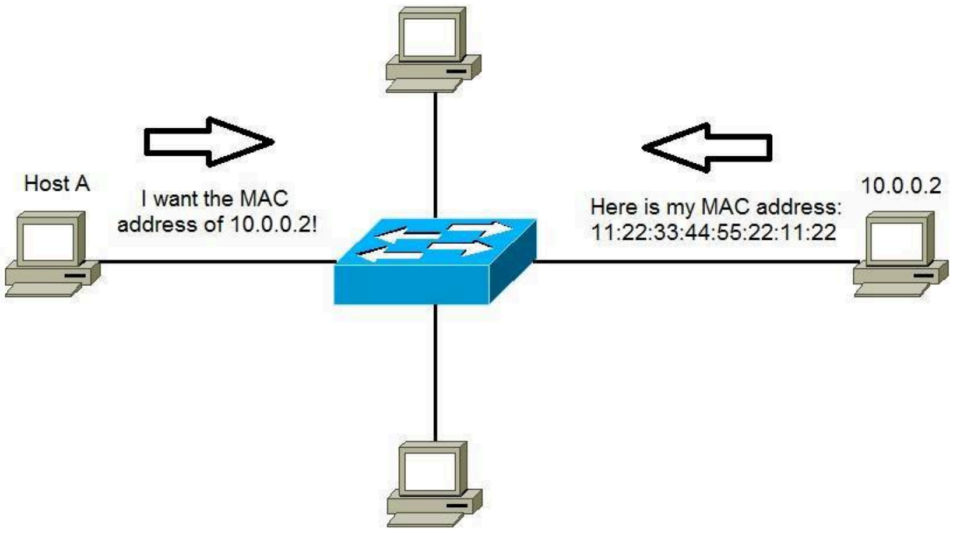
Address Resolution Protocol (ARP) is a protocol used to map an IP address to a MAC (Media Access Control) address within a local area network (LAN). Since data link layer communication (Ethernet) uses MAC addresses, ARP is essential for enabling devices to communicate over IP networks. Here's how ARP works:

Purpose of ARP:Resolve IP to MAC Address: When a device needs to send data to another device within the same network, it must know the destination's MAC address, which is used to deliver Ethernet frames. ARP helps map the destination IP address to its corresponding MAC address.

ARP Process:

1. **ARP Request**: If a device (e.g., PC1) wants to send data to another device but doesn't know the MAC address of the target device (e.g., IP 192.168.1.2), it sends an ARP request. The ARP request is a broadcast message, asking, "Who has IP address 192.168.1.2, and what is your MAC address?" The destination device (if it's on the same network) will recognize the IP and reply with its MAC address.
2. **ARP Response**: The device with the requested IP address (e.g., 192.168.1.2) responds by sending an ARP reply. This reply includes its MAC address, allowing the sender to associate the destination IP address with the corresponding MAC address.
3. **Caching**: After receiving the ARP reply, the sender stores the IP-MAC mapping in its ARP cache for a certain period of time, so it doesn't have to repeatedly broadcast ARP requests for the same device.

Types of ARP: Proxy ARP: A router responds on behalf of other devices.
Gratuitous ARP: A device announces or updates its own IP-MAC mapping.



ARP Table: Each device maintains an **ARP table** that holds the mapping of IP addresses to MAC addresses. This helps the device quickly find the MAC address when sending data, rather than having to perform an ARP request each time.

Example Scenario:



1. **PC1 wants to send data to PC2** on the same local network.
2. **PC1 knows PC2's IP address** (e.g., 192.168.1.2) but needs its MAC address.
3. **PC1 sends an ARP request** asking, “Who has 192.168.1.2? Please send me your MAC address.”
4. **PC2 responds with its MAC address**.
5. **PC1 sends the data** using the obtained MAC address.

ARP is essential for local network communication, enabling devices to send data frames using the correct MAC address associated with the destination IP. It improves network efficiency by caching mappings and reduces unnecessary ARP requests. However, ARP is vulnerable to **ARP spoofing**, where attackers can mislead devices by sending false MAC address information.

Reverse Address Resolution Protocol (RARP) is a network protocol that allows a device to obtain its IP address from a network server, given its MAC address. It is used primarily by devices that know their MAC address but do not have an IP address.

Key Points about RARP:

1. **Purpose**: It enables a machine to find its IP address by sending a request with its MAC address.
2. **Process**:
    - A machine broadcasts a RARP request to the network.
    - The **RARP server** responds with the corresponding IP address associated with the MAC address.
3. **Limitations**:
    - RARP only works within the same subnet because it relies on broadcast at the data link layer.
    - A separate RARP server is required for each subnet, making it less scalable.

    Due to these limitations, RARP has been largely replaced by more efficient protocols like **DHCP** and **BOOTP**, which provide dynamic IP address allocation and work across multiple subnets.

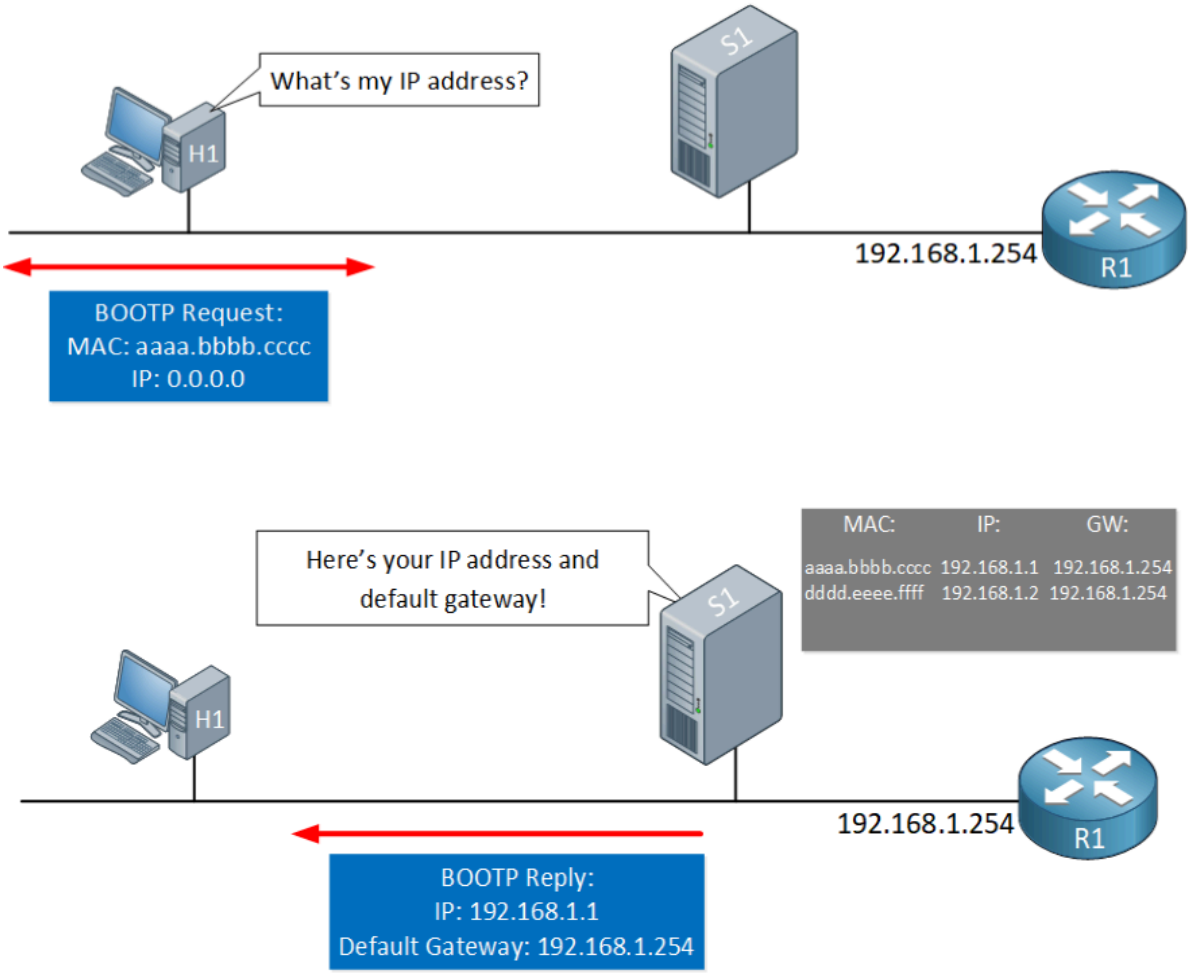


BOOTP (Bootstrap Protocol) is a network protocol that allows a device to automatically receive an IP address and boot its operating system without user intervention.

Key Points:

- **Automatic Configuration**: BOOTP assigns an IP address to a device and helps it start its OS by communicating with a BOOTP server.
- **Request and Response**: The client sends a broadcast request (UDP port 68 to port 67), and the server responds with an IP address.
- **Addressing**: The client does not know its own IP address or the server's, so it uses 0.0.0.0 as the source address and 255.255.255.255 as the destination for the broadcast.
- **Inter-Network Communication**: BOOTP works across different networks, unlike RARP, because both client and server are application-layer processes.
- **Broadcast Limitation**: Since BOOTP requests are broadcasts, they can't pass through routers. An intermediary device (called a **BOOTP relay agent**) is used to forward the request across networks.

BOOTP is now largely replaced by **DHCP**, which offers more flexibility and features for dynamic IP address allocation.



DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and other network settings to devices on a network.

Key Features:

1. **Dynamic IP Allocation**: DHCP gives devices a temporary IP address from a pool for a specific period.
2. **Static IP Allocation**: DHCP can assign a fixed IP address to a device based on its MAC address.
3. **Configuration Info**: DHCP can also provide settings like the gateway, DNS servers, and subnet mask.

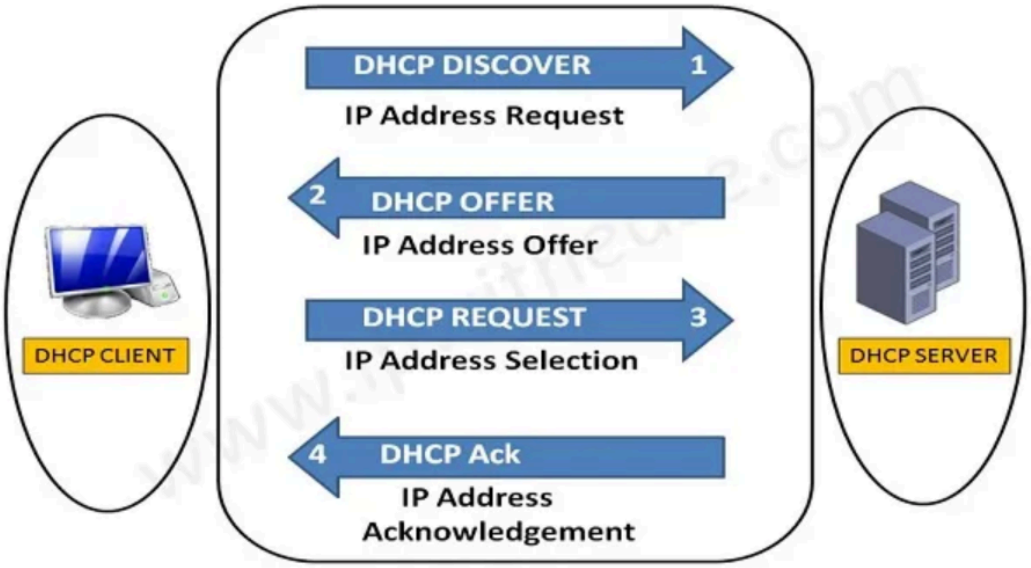
DHCP Process:

1. **Discover**: The device sends a request to find a DHCP server.
2. **Offer**: The server offers an available IP address and settings.
3. **Request**: The device accepts the offer.

4. **Acknowledge**: The server confirms and assigns the IP.

Benefits:

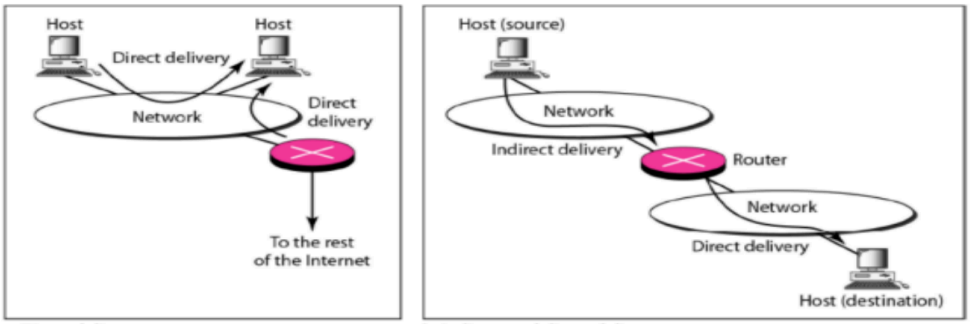
- **Automatic Configuration**: Devices get network settings without manual setup.
- **Efficient Address Management**: DHCP manages IP address distribution automatically, reducing conflicts.



Delivery, Forwarding, and Unicast Routing Protocols

1. Delivery: refers to the process of sending a data packet from the sender to the recipient.

- **Direct Delivery**: If the sender and the recipient are on the same network (subnet), the packet can be delivered directly without the need for a router. This is a simple case where the destination IP address corresponds directly to the device in the same local network.
- **Indirect Delivery**: When the sender and recipient are on different networks (i.e., the destination is not within the same subnet), the packet must be forwarded via one or more routers. These routers are responsible for forwarding the packet based on the routing information they hold.



a. Direct delivery

b. Indirect and direct delivery

2. Forwarding:

- **Forwarding** is the act of moving a packet along the network towards its destination by using a routing table.
- When a packet arrives at a router, the router needs to decide the best next hop (next router or destination). The router looks at its **routing table** to make this decision.

Forwarding Techniques:

1. **Next-Hop Method**:
    - In this method, the router's routing table only stores the address of the next hop (the next router or final destination) instead of the complete route. This helps reduce the size of the routing table and makes forwarding decisions faster and more efficient.
    - The next hop method is efficient because routers only need to know where to forward the packet next, not the entire path.

a. Routing tables based on route

Destination	Route
Host B	R1, R2, host B

Routing table for host A

Destination	Route
Host B	R2, host B

Routing table for R1

Destination	Route
Host B	Host B

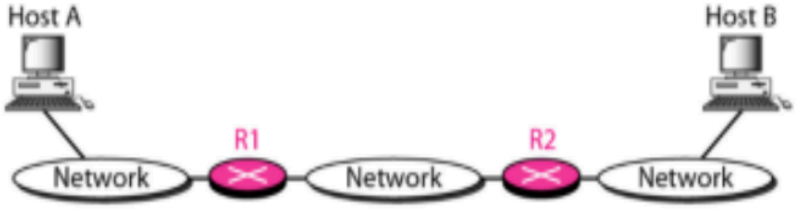
Routing table for R2

b. Routing tables based on next hop

Destination	Next hop
Host B	R1

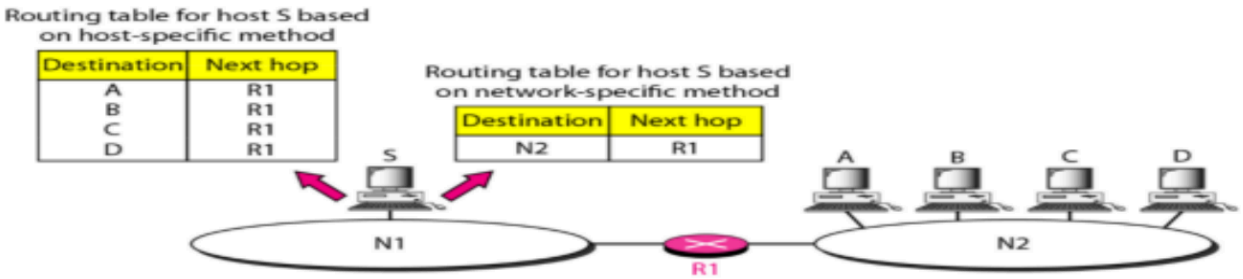
Destination	Next hop
Host B	R2

Destination	Next hop
Host B	...



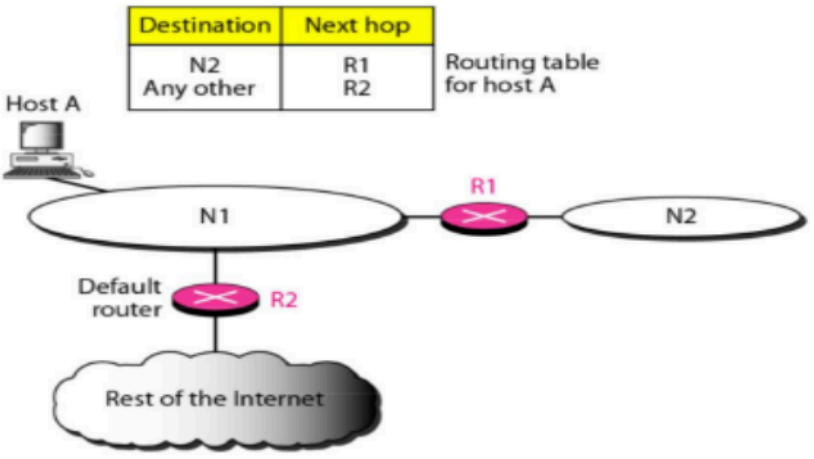
2. **Network-Specific Method**:
    - This method reduces the size of the routing table by listing entire networks, instead of individual host addresses. For instance, if there are 1,000 hosts on the same network, only one entry is needed for that network rather than 1,000 individual entries.
    - This method simplifies routing, especially in large networks.

Figure 22.3 *Host-specific versus network-specific method*



3. **Default Method**:
    - A default route is used when there is no specific match in the routing table for a destination. Instead of listing every possible network in the Internet, routers maintain a default route (often referred to as 0.0.0.0/0) to forward packets that don't match any other entry.
    - This allows a router to forward traffic to a "default" router for unknown destinations, streamlining the routing process.

Figure 22.4 *Default method*



Forwarding Process:When a packet arrives at a router, the router checks the destination address of the packet, then consults its routing table to determine the next-hop address. It then forwards the packet to the appropriate interface or next-hop router to continue its journey.

3. Routing Table:

Routing is the process of determining the best path for data packets to travel from their source to their destination across a network. It involves forwarding packets through intermediary devices such as routers, which examine the destination address and use routing tables to decide the next hop in the journey.

- **Routing Tables** are crucial for forwarding decisions. They store information about the paths that packets should take to reach their destination.
    - **Static Routing Table**:
        - These tables are manually configured and don't change unless an administrator updates them. They are simple and reliable for small, stable networks but are impractical for large or dynamic networks.
    - **Dynamic Routing Table**:
        - Dynamic tables are updated automatically using dynamic routing protocols like **RIP (Routing Information Protocol)**, **OSPF (Open Shortest Path First)**, and **BGP (Border Gateway Protocol)**.
        - These tables adjust automatically to changes in the network topology (e.g., if a link goes down or a router is added).

Routing Table Fields:

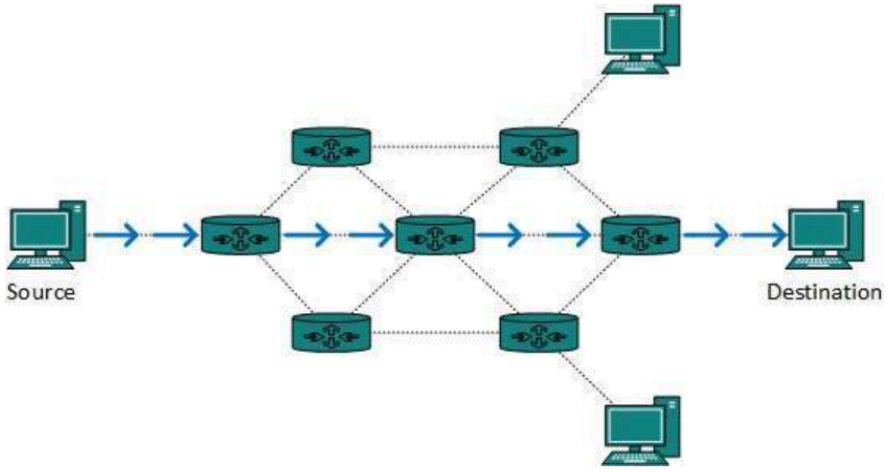
1. **Mask**: Indicates the subnet mask, which is used to determine if an IP address is within a given range.
2. **Network Address**: The destination network or the IP address range that the router can route to.
3. **Next-Hop Address**: The address of the next router that the packet should be forwarded to.
4. **Interface**: The router interface (or network card) through which the packet should be sent.
5. **Flags**: Used to indicate the status of the route (e.g., **U** for "up", **G** for "gateway", **D** for "dynamic").
6. **Reference Count**: The number of devices or routes that use this route.
7. **Use**: A counter indicating the number of packets forwarded via this route.

Figure 22.10 *Common fields in a routing table*

Mask	Network address	Next-hop address	Interface	Flags	Reference count	Use
.....	.....	.....	.....	.....	.....	.....

Unicast Routing Protocols:

- **Unicast Routing Protocols** are used to manage the paths packets take across a network, ensuring that data is correctly routed to its destination.



Intra-domain Routing (IGP):

- Manages routing within a single autonomous system (AS).
- **Examples:**
    - **RIP:** Uses hop count for routing.
    - **OSPF:** Uses link-state for efficient routing.

Inter-domain Routing (EGP):

- Manages routing between different ASs.
- **Example:**
    - **BGP:** Determines best paths across multiple ASs on the Internet.

Types of Unicast Routing Protocols:

1. **Distance Vector Routing Protocols**:
2. **Link State Routing Protocols**:
    - OSPF(Dikstra is used)

Distance Vector Routing Algorithm:

The **Distance Vector Routing Algorithm** is a distributed algorithm used by routers to determine the best path to send packets from a source to a destination. The algorithm is based on the concept of each router knowing only the distance (in terms of hops or cost) to each of its directly connected neighbors, and then sharing this information with them.

The key idea is that each router periodically sends out its routing table (called a "distance vector") to its immediate neighbors. Based on this information, each router updates its routing table to reflect the best known paths to all destinations.

How It Works:

1. **Each router maintains a routing table** that contains the distance (or cost) to all destinations (other routers or hosts in the network). Initially, each router only knows the distance to its immediate neighbors (usually set to 1 hop).
2. **Periodic updates**: Routers periodically send their routing table to their neighbors, and upon receiving the updated tables from neighbors, they update their own routing table by applying the **Bellman-Ford equation**:

   $$\mathrm{dx}(y) = \min_{v} \left(c(x, v) + \mathrm{dv}(y) \right)$$

   Where:
   - dx(y) is the least cost from router x to destination y.
   - c(x, v) is the cost of the link from router x to its neighbor v.
   - dv(y) is the cost to reach destination y from router v.
3. **Propagation**: The routers keep sending their updated distance vectors until all routers converge to the correct and stable routing tables.

   Example: https://www.youtube.com/watch?v=JMzbdR9iBFY

The count-to-infinity problem is a routing loop issue in Distance Vector Routing (DVR) protocols.

What is the Count-to-Infinity Problem?

In DVR, routers exchange routing information (distance vectors) with neighbors. When a link fails, a router may receive an updated distance vector from a neighbor, indicating an infinite distance (or a high metric) to the destination. However, if the neighbor also relies on the original router for reachability, an infinite loop occurs.

Example:

Router A -> Router B -> Router C

1. Router C fails.

2. Router B sends an update to Router A with infinite distance to Router C.

3. Router A sends an update to Router B with a higher distance to Router C.

4. Router B receives the update and increments the distance further.

5. This process repeats, causing the distance to increment indefinitely.

Consequences:

1. Routing loops: Packets are forwarded in loops, consuming network resources.

2. Network instability: Routing tables become inconsistent, leading to black holes.

Solutions:

1. Split Horizon: Routers do not send routing information back to the neighbor from which it was received.

2. Poison Reverse: Routers send an infinite distance to a destination that is unreachable, preventing further updates.

3. Triggered Updates: Routers send updates immediately when a change occurs, rather than waiting for periodic updates.

4. Hold-Down Timers: Routers delay updates for a specified time to prevent rapid changes.

5. Route Validation: Routers verify route information before updating their routing tables.

Modern Solutions:

1. Link-State Routing (LSR): Replaces DVR with a more robust, loop-free routing protocol.

2. Enhanced Interior Gateway Routing Protocol (EIGRP): Combines DVR with LSR features to prevent routing loops.

3. Border Gateway Protocol (BGP): Uses path vectors to prevent routing loops.

Prevention Techniques:

1. Design network topology carefully: Avoid redundant paths.

2. Implement routing protocol authentication: Ensure only authorized routers exchange routing information.

3. Monitor network performance: Detect and respond to routing loops quickly.

Shortest Path Routing in Unicast Routing

Unicast Routing refers to the process of sending data packets from one source to one specific destination. **Shortest Path Routing** is a key concept used in unicast routing to ensure that data travels through the most efficient path in terms of cost, which is typically measured by hop count or other metrics like bandwidth, latency, or link cost.

In unicast routing, the goal is to find the best path for a packet from the source to the destination node. **Shortest Path Routing** aims to minimize the path cost between the source and destination. This routing method is typically used in **Link-State** routing protocols (e.g., OSPF, IS-IS) and **Distance Vector** routing protocols (e.g., RIP), with the **Dijkstra Algorithm** being the most common algorithm used to determine the shortest path.

Dijkstra's Algorithm (Used in Link-State Routing): Dijkstra's algorithm is a famous algorithm used for finding the shortest path in a network. It works by calculating the shortest path from a source node to all other nodes in the network. The algorithm uses a greedy approach to explore the network.

Steps:

- **Initialization:** Set the distance to the source node as zero and the distance to all other nodes as infinity.
- **Selection of Node:** Choose the node with the smallest tentative distance.
- **Update Neighboring Nodes:** For the current node, examine its unvisited neighbors and calculate their tentative distance from the source.
- **Repeat:** Mark the current node as visited and repeat the process for the next unvisited node with the smallest tentative distance, until all nodes are visited.

EX:https://www.youtube.com/watch?v=p9fPBS2S_Vk&t=1s

EX:https://www.youtube.com/watch?v=eXXM7ukL4_k

In OSPF (Open Shortest Path First), there are four types of network links classified based on the network topology and the role of the routers in the OSPF domain. The classification is based on the **OSPF network topology** and the way routers interact with each other. These four types are:

1. **Point-to-Point (P2P) Link**
    - **Description**: Direct connection between two routers.
    - **Example**: Leased line between two routers.
2. **Broadcast Link**
    - **Description**: A network where multiple routers are connected, like Ethernet.
    - **Example**: Ethernet network.
    - **Key Point**: Elects a Designated Router (DR) to manage communication.
3. **Non-Broadcast Multi-Access (NBMA) Link**
    - **Description**: A multi-router network that doesn't support broadcast (e.g., Frame Relay).
    - **Example**: Frame Relay network.
    - **Key Point**: Needs manual configuration for neighbors.
4. **Point-to-Multipoint (P2MP) Link**
    - **Description**: One router connects to multiple others, but each connection is treated as point-to-point.
    - **Example**: Hub-and-spoke network.

These link types affect how OSPF routers communicate and share routing information.

What is Flooding?

Flooding refers to the process of sending a message to all nodes (routers, hosts) in the network. When a router receives a packet, it forwards the packet to all of its neighbors, except the one from which it received the packet. This continues until the packet reaches all nodes in the network.